

Memory Circuits & Systems

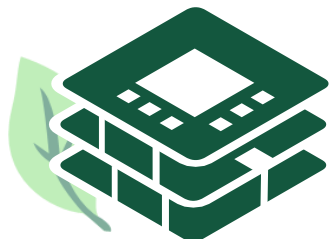
Final Project

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SCALE Lab

Sustainable Circuits and
Architectures for layered
Eco-chips Lab

NYCU

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YANG MING CHIAO TUNG
UNIVERSITY

■ 25% Final Project

- ◆ 2-4 students as a group
- ◆ Final report and presentation (due date: 6/30)

■ 2kb SRAM

- ◆ SRAM array (Post-sim is required)
- ◆ Column-based peripheral circuits (Post-sim is required)
- ◆ Row-based peripheral circuits (post-sim is optional)
- ◆ Global control circuits (post-sim is optional)
- ◆ Tracking circuits under PVT condition (optional)

■ Open Topics related to CIM or Memory Sub-System

- ◆ Submit your proposal and expected results by email before 6/7

Spec of 128 x 16 (2kb) SRAM Design



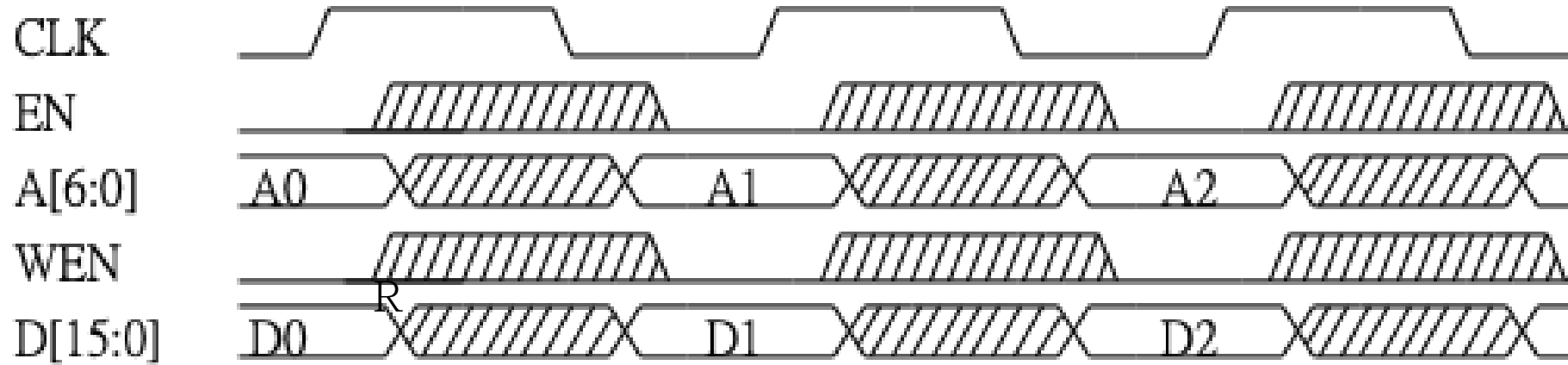
- The size of SRAM in the project is 2kb by 128x16.
- ◆ The minimum frequency of this SRAM is 1GHz.
- ◆ All the input data have to pass one smallest buffer to prevent infinite driving ability.
- ◆ The rise time and fall time of all inputs are 50ps respectively.

I/O description for SRAM

Pin Name	I/O	Description
Clock	Input	Clock input
A[6:0]	Input	Address
D[15:0]	Input	Data input
WEN	Input	Write Enable (1 for write, 0 for read)
Q[15:0]	Output	Data output

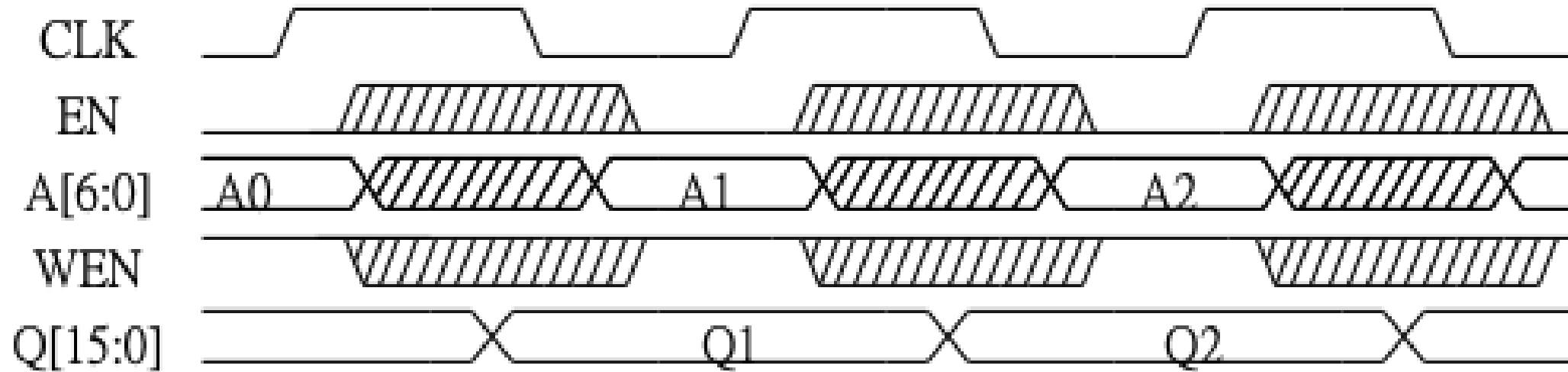
Waveform of Write Operations

Waveform of write operations



- The duty cycle for the input clock is free for your design.
- WEN is set to low and high for write and read operations, respectively.

Waveform of Read Operations



- WEN is set to low and high for write and read operations, respectively.
- A[6:1] are for the row decoder, and A[0] is for the column decoder.
- For read operations, Q[15:0] have to be valid before the next rising/falling edge for the functionality.